

# Simulation-based Time Series Analysis:

## Likelihood-free Estimation, Testing for Structure, and Prediction with Transformers using Synthetic Data

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# Overview of presentation

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## Part I

### Testing

Economics application

Hardness result

Physical dependence

Type I error control

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## Part III

### Prediction

Foundation models

Sim-and-regression

Conf. distributions

Calibration

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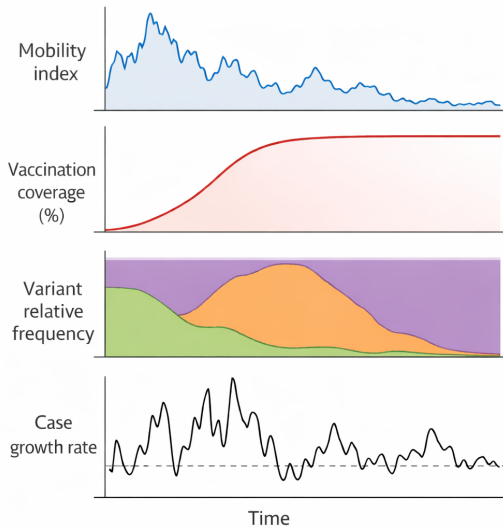
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- ▶ Example: time series in epidemiology

# COVID-19 time series May 2020–2022 ( $n=365 \times 2=730$ , $d=4$ )



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- ▶ Example: links in the global economic system

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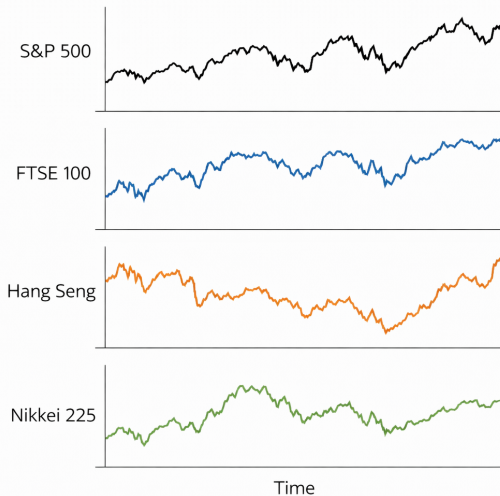
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- ▶ We show a subset of the results based on BH-adjusted p-values

# Closing prices of each stock index over time



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- ▶ We reject every null hypothesis at the significance level  $\alpha = 0.05$



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- ▶ Next, we present the testing procedure

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- **Inputs:** Data  $X_{1:n}$ ,  $Y_{1:n}$ ,  $Z_{1:n}$ , significance level  $\alpha$ , number of simulations  $s$ ,  $p \in [2, \infty]$  for norm in test statistic, and method for selecting the window size  $L$

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- ▶ Calculate test statistic on time series of residual products

$$T(\hat{R}_{1:n}) = \max_{j \in [n]} \left\| \frac{1}{\sqrt{n}} \sum_{t \leq j} \hat{R}_t \right\|_p$$

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- ▶ Reject null hypothesis at level  $\alpha$  if  $T(\hat{R}_{1:n}) > \hat{q}_{1-\alpha}^{\text{boot}}$ , else retain

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- ▶ **Implication:** We can only hope to achieve Type I error control on a subset of null distributions  $\mathcal{P}_n^{\text{CI}}$ , so must restrict the null hypothesis to make CI testing feasible
- ▶ **Key assumption:** uniform decay of physical dependence measure (Wu 2005) for all time series  $X_{1:n}$ ,  $Y_{1:n}$ , and  $Z_{1:n}$

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- ▶ Let  $\Theta$  be a parameter space, which can be infinite-dimensional
- ▶ Let  $G_t^{(n)} : \mathbb{R}^\infty \times \Theta \rightarrow \mathbb{R}^d$  be some measurable function,  $t, n, d \in \mathbb{N}$

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- ▶ Let  $P_{\theta,n}$  be the (finite-dimensional) distribution of  $(G_t^{(n)}(\epsilon_t, \theta))_{t \in [n]}$
- ▶ Let  $P_\theta$  be the law of  $(G_t(\epsilon_t, \theta))_{t \in \mathbb{N}}$  for some limiting mapping  $G_t$
- ▶ Denote the collections by  $\mathcal{P}_n = \{P_{\theta,n} : \theta \in \Theta\}$  and  $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$



## Key assumption: Decaying influence of noise inputs (Wu 2005)

- Let  $(\epsilon_i^*)_{i \in \mathbb{Z}}$  be an iid copy of  $(\epsilon_i)_{i \in \mathbb{Z}}$ . For  $\ell \in \mathbb{Z}$ ,  $j \in \mathbb{N}_0$ , denote

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### Assumption

*There exist  $\Psi, \rho > 0$  such that, for some  $q > 2$  and all  $j \in \mathbb{N}$ , we have*

$$\sup_{\theta \in \Theta} \sup_{\ell \in \mathbb{Z}} \sup_{i \in \mathbb{N}} \sup_{t \in \mathbb{N}} \left\| G_t^{(i)}(\epsilon_\ell, \theta) - G_t^{(i)}(\tilde{\epsilon}_{\ell,j}, \theta) \right\|_{L^q(\theta)} \leq \Psi(j \vee 1)^{-\rho}$$

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where  $\|\cdot\|_{L^q(\theta)} = \mathbb{E}_\theta (\|\cdot\|_2^q)^{1/q}$ ,  $\mathbb{E}_\theta(\cdot)$  is expectation w.r.t.  $\theta$ -dependent distribution

## We prove our test has uniformly asymptotic Type I error control

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### Theorem (Informal)

Recall the test statistic  $T(\hat{R}_{1:n})$  and estimated quantile  $\hat{q}_{1-\alpha}^{\text{boot}}$ . We have

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_n^*} \mathbb{P}_P \left( T(\hat{R}_{1:n}) > \hat{q}_{1-\alpha}^{\text{boot}} \right) \leq \alpha.$$

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- ▶ **Key assumption:** We know  $G_t^{(n)}$  and can simulate from it at any value of  $\theta \in \Theta$

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- ▶ For nonstationary time series, need rolling-window estimator (not discussed here)

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  - ▶ Extract early layer of neural network

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- ▶ Can we just use *random features* of the data instead?
- ▶ No need to craft features, computationally efficient, little knowledge needed

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- ▶ Probabilistic embedding theorem due to Sauer, J. A. Yorke, and Casdagli (1991)
- ▶ For “almost every”  $C^1$  smooth function  $\Phi$  from  $\mathbb{R}^p$  to  $\mathbb{R}^{2p+1}$ :
  1.  $\Phi$  **is one-to-one** on compact subsets of  $\mathbb{R}^p$
  2.  $\Phi$  **has a nice derivative** on certain compact subsets of  $\mathbb{R}^p$ 
    - Details: the  $(2p + 1) \times p$  Jacobian matrix of  $\Phi$  has full rank  $p$  on each compact subset of a smooth manifold contained in  $\mathbb{R}^p$

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- ▶ If physical dependence measure (Wu 2005) decays uniformly over  $\Theta$ , we can estimate  $\Phi(\theta)$ ,  $\theta \in \Theta$ , via time-averages of  $2p + 1$  random features

$$\frac{1}{n-m} \sum_{t=m+1}^n \varphi \left( [X_t(\theta), \dots, X_{t-m}(\theta)]^\top \right)$$

## Example: Mean of Gaussian

- **Goal:** Estimate  $\theta_0 = 0$  given  $X_1^{\text{obs}}, \dots, X_n^{\text{obs}} \stackrel{\text{iid}}{\sim} \mathcal{N}(\theta_0, 1)$  with  $\Theta = [-3, 3]$

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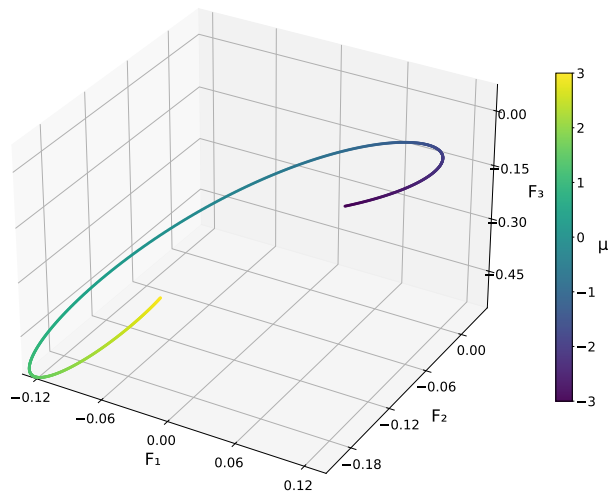
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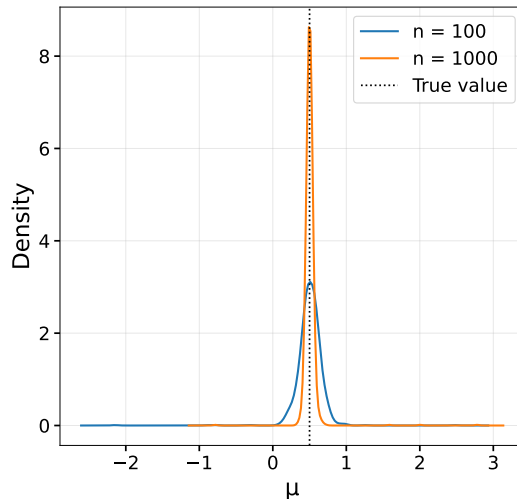
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- ▶ We will compare the MSE of our estimator  $\hat{\theta}$  with  $\hat{\theta}^{\text{MLE}} = \frac{1}{n} \sum_{t=1}^n X_t$

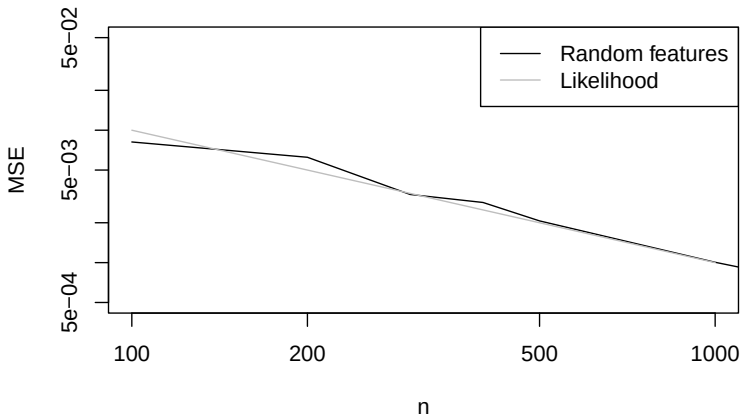
# Time-average of 3 features, colored by parameter value ( $n = 1000$ , $s = 10$ )



# Density plot based on 1,000 independent estimates



# Theoretical MSE of MLE vs MSE of our estimator (100 replicates per $n$ )



## Theoretical guarantees for two estimators

- We prove consistency  $\hat{\theta} \xrightarrow{P} \theta_0$  and asymptotic normality

$$\sqrt{n}(\hat{\theta} - \theta_0) \Rightarrow N\left(0, (\Gamma^\top \Gamma)^{-1} \Gamma^\top V \Gamma (\Gamma^\top \Gamma)^{-1}\right)$$

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- ▶ We also prove the same guarantees for a rolling-window estimator



## Theoretical guarantees for two estimators

- ▶ We prove consistency  $\hat{\theta} \xrightarrow{P} \theta_0$  and asymptotic normality

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- ▶ We also prove the same guarantees for a rolling-window estimator
- ▶ Examples of nonstationary time series:
  - ODEs with *temporally dependent* observational noise
  - State-space models with trend and seasonality

## Paradigm shift in prediction: Time series foundation models (TSFMs)

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- ▶ Statistical foundations are unclear, especially for nonstationary time series
- ▶ We develop theory, and propose *simulation-and-regression* methods
- ▶ In the context of state-space modeling:
  - **Smoothing:** estimate past latent states  $X$  with current info
  - **Forecasting:** predict future observations  $Y$  or latent states  $X$  with current info

## Algorithm for classical simulation-and-regression in seq2seq setting

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- ▶ Step 5: Apply the estimated function  $\hat{f}$  to predict the latent states

$$\hat{X} = \hat{f}(Y^{\text{obs}})$$

## Accounting for parameter uncertainty

- ▶ The previous method aims to minimize the Monte Carlo approximation to

$$R_{P_{\hat{\theta}}}(f) = \mathbb{E}_{P_{\hat{\theta}}} \left[ \sum_{t \in \mathcal{T}} L(X_t - [f(Y)]_t) \right]$$

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- ▶ Why not account for parameter uncertainty like our Bayesian friends?

## Our method: Simulation-and-regression with parameter uncertainty

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- ▶ Theory of confidence distributions from D. Liu, R. Y. Liu, and Xie (2022)
- ▶ We propose minimizing the Monte Carlo approximation to

$$R_{P_{\widehat{\text{mix}}}}(f) = \mathbb{E}_{\theta \sim \widehat{\text{CD}}_n} \left[ \mathbb{E}_{P_\theta} \left[ \sum_{t \in \mathcal{T}} L(X_t - [f(Y)]_t) \right] \right] = \mathbb{E}_{P_{\widehat{\text{mix}}}} \left[ \sum_{t \in \mathcal{T}} L(X_t - [f(Y)]_t) \right]$$

for some loss  $L$ , where the expectation is w.r.t. the mixture distribution  $P_{\widehat{\text{mix}}} = \int_{\Theta} P_\theta d\widehat{\text{CD}}_n(\theta)$  of sequences  $X, Y$  corresponding to  $\widehat{\text{CD}}_n$

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- ▶ When  $\widehat{\text{CD}}_n$  is Dirac delta at  $\hat{\theta}$ , this is like classical simulation-and-regression
- ▶ We propose a few possible directions to pursue

## Questions we aim to answer

- ▶ **Calibration:** Does accounting for parameter uncertainty improve the calibration of pre-trained/untrained transformers that output multiple conditional quantiles?

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- ▶ **Pre-training:** When does a pre-trained transformer fine-tuned with our method outperform a transformer trained from scratch with our method?
- ▶ **Improving fine-tuning and simulation-and-regression:** When does accounting for parameter uncertainty improve methods based on parameter point estimates?



# Thank you!

Questions or comments?

You can reach me at:  
`mwiecksosa@cmu.edu`

# How to simulate to get pre-training data for forecasting with TSFMs

- For forecasting with TSFMs, Dooley et al. (2023) suggest training on simulations from *multiplicative seasonality-trend-noise decomposition models* (many  $\theta \in \Theta$ )
- We consider an extension with multiplicative autoregressive noise

$$Y_t = \text{Se} \left( t, \theta^{\text{Se}} \right) \text{Tr} \left( t, \theta^{\text{Tr}} \right) \exp \left( \kappa X_t - \frac{\kappa^2 \sigma_X^2}{2(1 - \phi^2)} \right) \exp \left( \varepsilon_t^Y - \frac{\sigma_Y^2}{2} \right)$$
$$X_t = \phi X_{t-1} + \varepsilon_t^X$$

for some scaling factor  $\kappa > 0$  and autoregressive coefficient  $\phi \in (-1, 1)$ , with  $X_0 \sim N(0, \frac{\sigma_X^2}{1 - \phi^2})$  and  $\varepsilon_t^X \stackrel{\text{iid}}{\sim} N(0, \sigma_X^2)$ ,  $\varepsilon_t^Y \stackrel{\text{iid}}{\sim} N(0, \sigma_Y^2)$  for some  $\sigma_X, \sigma_Y > 0$

- Seasonality: linear combination of sines, cosines (weekly, monthly, yearly periods)
- Trend: product of linear trend and exponential trend
- Our rolling-window estimator can be used (consistent and asymptotically normal)

# Does accounting for parameter uncertainty improve the calibration of pre-trained/untrained transformers that output conditional quantiles?

- For untrained transformers, see answer to question: “In the context of simulation-and-regression, does accounting for parameter uncertainty achieve better performance than using parameter estimate?”
- For pre-trained transformers, see answer to question: “When does a pre-trained transformer fine-tuned with our method outperform a transformer trained from scratch with our method?”

In the context of simulation-and-regression, does accounting for parameter uncertainty achieve better performance than using parameter estimate?

– When does this inequality hold:

$$\mathbb{E}_{P_{\theta_0}} \left[ \sum_{t \in \mathcal{T}} L \left( X_t - [f_{\widehat{P}_{\text{mix}}}^*(Y)]_t \right) \right] < \mathbb{E}_{P_{\theta_0}} \left[ \sum_{t \in \mathcal{T}} L \left( X_t - [f_{P_{\hat{\theta}}}^*(Y)]_t \right) \right]$$

where both expectations are with respect to the distribution  $P_{\theta_0}$  of the sequences corresponding to the unknown true parameter  $\theta_0$ ,  $f_{\widehat{P}_{\text{mix}}}^*$  is the minimizer in the function class of  $R_{\widehat{P}_{\text{mix}}}(f)$ , and  $f_{P_{\hat{\theta}}}^*$  is the minimizer in the function class of  $R_{P_{\hat{\theta}}}(f)$

# When does a pre-trained transformer fine-tuned with our method outperform a transformer trained from scratch with our method?

- Intuitively, this occurs when the following conditions hold
- Large parameter uncertainty, e.g., when the sample size of the observed time series is small relative to the noise
- The distributions used in the pre-training phase are close to the true distribution, e.g., when we have domain knowledge or we have previously observed time series with similar distributions

## What do we mean by “almost every” function? Prevalent subset

### Definition (J. Yorke and Ott (2005))

Let  $V$  be a completely metrizable topological vector space. A Borel set  $E \subset V$  is said to be **prevalent** if there exists some Borel measure  $\mu$  on  $V$  such that:

1.  $0 < \mu(C) < \infty$  for some compact subset  $C$  of  $V$ .
2. The set  $E + v$  has full  $\mu$ -measure (that is, the complement of  $E + v$  has measure zero) for all  $v \in V$ .

– In this literature, it is common to say that, if  $E \subset V$  contains a prevalent Borel set, then “almost every” element of  $V$  lies in  $E$ . In this sense, “almost every”  $C^1$  smooth map from  $\mathbb{R}^p$  to  $\mathbb{R}^{2p+1}$  has the properties from the embedding theorem of Sauer, J. A. Yorke, and Casdagli (1991).

# What is a smooth manifold? (University of Toronto Lecture Notes) I

- A  $C^\infty$ -manifold is a Hausdorff topological space  $M$ , with countable basis, together with a  $C^\infty$ -atlas.
- An  $n$ -dimensional manifold is a space that locally looks like  $\mathbb{R}^n$ . To give a precise meaning to this idea, our space first of all has to come equipped with some topology (so that the word “local” makes sense).

## What is a smooth manifold? (University of Toronto Lecture Notes) II

- Recall that a *topological space* is a set  $M$ , together with a collection of subsets of  $M$ , called *open subsets*, satisfying the following three axioms: (i) the empty set  $\emptyset$  and the space  $M$  itself are both open, (ii) the intersection of any finite collection of open subsets is open, (iii) the union of any collection of open subsets is open.
- The collection of open subsets of  $M$  is also called the topology of  $M$ . A map  $f : M_1 \rightarrow M_2$  between topological spaces is called *continuous* if the pre-image of any open subset in  $M_2$  is open in  $M_1$ . A continuous map with a continuous inverse is called a *homeomorphism*.



## What is a smooth manifold? (University of Toronto Lecture Notes) III

- One ingredient in the definition of a manifold is that our topological space comes equipped with a covering by open sets which are homeomorphic to open subsets of  $\mathbb{R}^n$ .
- Let  $M$  be a topological space. An  $n$ -dimensional *chart* for  $M$  is a pair  $(U, \phi)$  consisting of an open subset  $U \subset M$  and a continuous map  $\phi : U \rightarrow \mathbb{R}^n$  such that  $\phi$  is a homeomorphism onto its image  $\phi(U)$ . Two such charts  $(U_\alpha, \phi_\alpha)$ ,  $(U_\beta, \phi_\beta)$  are  $C^\infty$ -compatible if the transition map

$$\phi_\beta \circ \phi_\alpha^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \rightarrow \phi_\beta(U_\alpha \cap U_\beta)$$

is a diffeomorphism (a smooth map with smooth inverse). A covering  $\mathcal{A} = (U_\alpha)_{\alpha \in A}$  of  $M$  by pairwise  $C^\infty$ -compatible charts is called a  $C^\infty$ -atlas.

## What is a smooth manifold? (University of Toronto Lecture Notes) IV

– We recall that a topological space is called *Hausdorff* if any two points have disjoint open neighborhoods. A *basis* for a topological space  $M$  is a collection  $\mathcal{B}$  of open subsets of  $M$  such that every open subset of  $M$  is a union of open subsets in the collection  $\mathcal{B}$ . For example, the collection of open balls  $B_\varepsilon(x)$  in  $\mathbb{R}^n$  define a basis. But one already has a basis if one takes only all balls  $B_\varepsilon(x)$  with  $x \in \mathbb{Q}^n$  and  $\varepsilon \in \mathbb{Q}_{>0}$ ; this defines a countable basis. A topological space with countable basis is also called *second countable*.

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